

INSPIRE WP4 - Health economics

Analysis update

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Why?

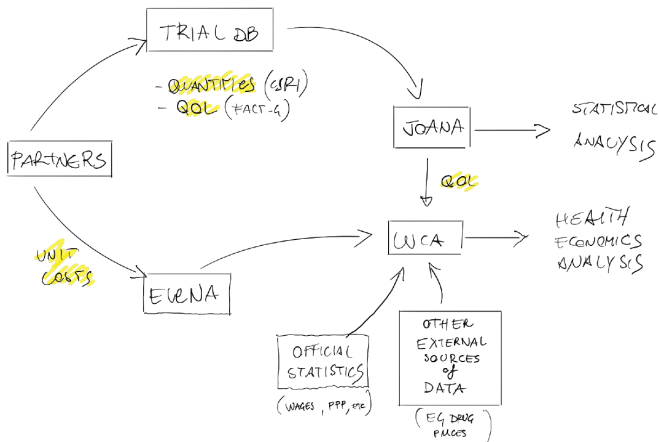
- **we are here:** from call HORIZON-HLTH-2021-DISEASE-04-01,

*[...] proposals under this topic should aim for delivering results that are directed, tailored towards and contributing to [... among other...] **reduce societal, healthcare and economic burden** associated with increasing demands of supportive, palliative, survivorship or end-of-life care services that is beneficial for citizens and preserves the sustainability of the health care systems.*

- **I am here:**
 - **background:** MSc economics (in a past life), 1 year post-graduate master in statistics, almost MSc statistics
 - Project initially financed with Dr. Massimo Costantini (I didn't participate in the first iterations/design); support for this part was needed, since formally assigned to AUSL-RE.

HE analysis - the overall picture I

Data collection for economics analysis performed alongside a clinical trial is less “self-contained” if compared to statistical analysis. The picture below depicts the data flow



HE analysis - the overall picture II

Main components (data) are three:

- 1 unit *costs*
- 2 *quantities*
- 3 *QOL/Utilities*

Three above are the **sections** of these slides too, where I'll give an update, discuss difficulties and summarize decisions taken (up to now).

In general, **approach** adopted:

- MVP (minimum viable product): results presented are *temporary* (both in code and data - upd jan 2026)
- make it run, make it right, ~~make it fast~~ , ...

Part I

Unit costs

Timeline I

2025-06-24: email from Elena - request for unit cost collection.

Home help / care worker / health care assistant			
A	B	C	D
Country: _____			
Institution: _____			
Currency to be considered for costs filled in below (eg: Euro): _____			
Main/reference year for cost filled in below (eg: 2025): _____			
Service	Mean Comprehensive Cost	Metric/Unit	
1. Hospital services			
Hospital inpatient ward		per day	
Intensive/Critical care ward/unit		per day	
Specialist/rehabilitation ward/unit		per day	
1. Other residential services			
Hospice/Palliative Care in-patient service		per day	
Nursing or residential home		per day	
2. Outpatient hospital services			
Emergency ambulance service		per visit	

Timeline II

july-november 2025: unit costs collected from partners
january-february 2026: normalization

Institution: <u>Charles university</u>											
Currency to be considered for costs filled in below* <u>CZK</u>											
Main/reference year for cost filled in below (eq: 20*) <u>2025</u>											
Imported data											
Service	value used	imputed	currency	year	per	source	notes	Mean cost	Metric/Unit	Source	Note/Comments
1. Hospital services											
Hospital inpatient ward	561	0	CZK	2025	day			561	per day	Czech ministry of health-	https://mzd.gov.cz/v
Intensive/Critical care ward/unit	4400	0	CZK	2025	day			4400	per day	same	there is a huge di
Specialist/rehabilitation ward/unit	2850	0	CZK	2025	day			2850	per day	same	
1. Other residential services											
Hospice/Palliative Care in-patient service	2400	0	CZK	2025	day			2400	per day	same	
Nursing or residential home	1500	0	CZK	2025	day			1500	per day	same	
2. Outpatient hospital services											
Emergency ambulance service	12000	1	CZK	2025	visit	FOR CH*		6000	paramedi	per visit	https://www.zzshmp.cz/cerf July covered for i
Emergency department (A&E Dept., Accident and *	5500	1	CZK	2025	visit	FOR CH*		1000 - 10000	per visit		https://mzd.gov.cz/wp-cont Depends on pres
Exercise rehabilitation at the hospital (i.e. pulmonar	560	0	CZK	2025	visit	FOR CH*		560	per visit		https://www.infoprotekare.cz This is the price c
Specialist medical outpatient visit (e.g. cancer, specialist palliative,									per visit		https://mzd.gov.cz/wp-content/uploads/2024/
Other non-medical hospital outpatient visit								x	per visit	x	
Day hospital department (e.g., chemotherapy and *	500	0	CZK	2025	visit	FOR CH*		Depends on to	per visit		https://mzd.gov.cz/wp-cont If there is radiothe
3. Community-based services											
General Practitioner/Primary Care Physician	1000	0	CZK	2025	visit			1000	per visit	https://szv.mzcr.cz/Nykon/	GPs are paid by

Unit cost normalization: some general rules

Standard year

The vast majority of unit costs refer to 2025, so this year was chosen as the base year. Costs from other years (e.g., some from Italy and France) were converted to 2025 using the project's long-term inflation index (NICE's 3.5%).

Multiple unit costs for same item

For a given unit cost, if *several values* were provided, we had to choose one (CSRI does not contain enough detail to determine which to use, patient won't know).

As a general rule, a simple *mean* was used at first instance, to get the "order of magnitude".

The sole exception is for *medications costs*, where to get the long term unit cost we'll adopt the lowest applicable amount.

3) Community based services (unit of reference) I

In all countries, unit costs for **community based services** (eg general practitioner, nurse etc) were provided on a **per visit** basis, as requested. The only exception was Italy where the available data were:

- **monthly** cost for GP;
- **hourly** cost for other professionals

Per-visit mean length (used to calculate unit cost) was estimated internally (by making assumptions and/or consulting colleagues):

- GP: assumed 20 hours of visits per week, 15 minutes each (as in BetterB).
- 60 min visits: Physiotherapist, Occupational therapist, Speech & language therapist, Psychologist
- 30 min visits: General practice nurse, District/Public health/Community nurse, Dietitian, Psychiatrist, Rehabilitation physician/Physiatrist, Specialist palliative care doctor/consultant, Specialist palliative care nurse, Home palliative care / home hospice care service, Social worker, Home help / care worker / health care assistant

4) Informal care: wages I

Regarding informal care:

- hours will be valued using the *mean wage* when adopting the societal perspective;
- *minimum wage* and *wages for unskilled workers* will not be considered due to limited availability

Mean wages were imputed for *all countries* to obtain fully available and consistent estimates across the dataset:

- *average full-time wage* is available from several sources
- *average weekly worktime* estimates are available overall and by sex : for now I used the mean number of hours for *males*, to better approximate full-time weekly hours (this still slightly underestimates it, but aligns better with the average full-time wage mentioned above).

4) Informal care: wages II

A	B	C	D	E	F	G	H
country_id	currency	average_net	year_net	average_hours_all	average_hours_male	average_hours_female	mean_hourly_wage
cz	CZK	40942	2025	34.5	36.5	32.1	261.7
dk	DKK	34849	2024	28.9	31.4	26.2	259.0
fr	EUR	2733	2024	30.9	33.4	28.2	19.1
it_re	EUR	2017	2024	33.9	36.9	29.8	12.8
it_mi	EUR	2017	2024	33.9	36.9	29.8	12.8
no	NOK	41649	2024	26.6	28.7	24.3	339.2
uk_en	GBP	2374	2025	31.2	34.5	27.6	16.1
uk_sc	GBP	2374	2025	31.2	34.5	27.6	16.1

5) Equipment: purchase vs rent I

For some equipment purchases (not rentals), purchase costs were available. These were converted to an equivalent monthly rental by assuming 5 years of use and discounting 60 equal monthly payments so their present value equals the purchase cost

$$\text{Purchase cost} = \frac{\text{MR}}{(1+i)^{1/12}} + \frac{\text{MR}}{(1+i)^{2/12}} + \dots + \frac{\text{MR}}{(1+i)^{60/12}}$$

With $i = 0.035$ (annual project interest rate, 3.5% as above) and a known purchase cost, the only unknown is the monthly equivalent rent (MR), which can be found by:

$$\text{MR} = \frac{\text{Purchase cost}}{\left[\frac{1}{(1+i)^{1/12}} + \frac{1}{(1+i)^{2/12}} + \dots + \frac{1}{(1+i)^{60/12}} \right]}$$

5) Equipment: purchase vs rent II

<u>cost_id</u>	<u>cost_description</u>	<u>country_id</u>	<u>purchase_cost</u>	<u>currency</u>	<u>year</u>	<u>monthly_rent</u>
505	Walking stick	fr	12.7	EUR	2025	0.2
501	Ambulator oxygen	no	140.1	NOK	2025	2.5
502	Long term oxygen therapy	no	4102.0	NOK	2025	74.5
503	Non-invasive ventilation	no	3208.0	NOK	2025	58.3
508	Feeding pump	no	3500.0	NOK	2025	63.6

Other rules I

Other special rules

it_mi inherited unit costs from it_re, uk_sc from uk_en (only where needed)

Currency and purchase power conversion

To adjust for different costs of living (all in 2025 EUR), purchasing power parity was applied using the same methodology as BETTER-B and Diadic.

Missing unit costs imputation

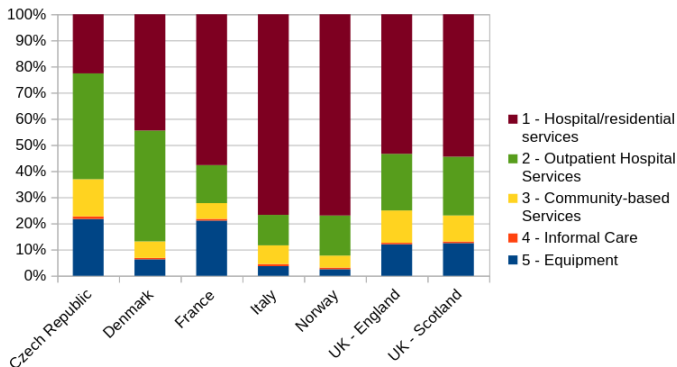
Similarly to BETTER-B/Diadic, this was done using unit-costs for same good/service in other countries (and normalizing for nationwide relative prices).

Final unit costs (EUR 2025)

description	<u>cz</u>	dk	fr	<u>it_mi</u>	<u>it_re</u>	no	<u>uk_en</u>	<u>uk_sc</u>
Hospital inpatient ward	27.7	819.3	1541.6	683.9	683.9	1110.3	659.4	659.4
Intensive/Critical care ward	217.3	2560.4	938.0	2082.9	2082.9	5759.9	2109.7	2109.7
Specialist/rehab ward/unit	140.8	819.3	592.3	650.3	650.3	1596.1	620.9	620.9
Hospice	118.5	372.1	451.0	248.9	248.9	1318.5	505.4	399.2
Nursing or residential home	74.1	164.0	62.4	135.0	135.0	203.3	174.7	215.1
Emergency ambulance	592.6	189.1	77.5	87.0	87.0	234.4	288.4	288.4
Emergency dept (A&E)	271.6	1194.2	349.1	161.4	161.4	329.6	124.9	124.9
Hospital rehab	27.7	56.0	98.6	13.4	13.4	69.4	60.1	60.1
Specialist	64.7	876.0	25.2	24.9	24.9	208.2	206.7	206.7
Other outpatient visit	53.0	137.5	97.4	24.9	24.9	208.2	147.5	147.5
Day hospital	24.7	2063.0	254.7	265.1	265.1	936.9	823.8	823.8
General Practitioner	49.4	17.3	28.5	32.1	32.1	49.9	43.2	43.2
General practice nurse	11.0	17.3	20.3	23.5	23.5	35.5	13.2	33.7
District/community nurse	11.0	28.6	20.3	23.5	23.5	35.5	20.7	33.7
Physiotherapist	27.7	53.1	19.5	13.5	13.5	34.1	25.4	33.7
Occupational therapist	27.7	23.6	16.7	13.0	13.0	29.3	25.4	33.7
Speech & language therapist	9.9	23.6	16.7	13.5	13.5	29.3	25.4	33.7

Final unit costs (sum by section, EUR 2025)

Category	Czech Republic	Denmark	France	Italy	Norway	UK - England	UK - Scotland
1 - Hospital/residential services	578	4735	3585	3801	9988	4070	4004
2 - Outpatient Hospital Services	1034	4516	902	577	1987	1651	1651
3 - Community-based Services	364	671	374	355	597	944	735
4 - Informal Care	25	59	40	40	79	43	43
5 - Equipment	555	669	1311	181	325	918	918



Part II

CSRI and quantities

Missingness handling I

General considerations:

- missingness, in a sense, is worse for efficiency than the most infamous virus: anything that is “touched” by a missing value becomes missing immediately with 100% probability;
- especially problematic with “composite outcomes” as here: a pity if quantities are all available but just one (which makes prices · quantities all missing)
- the best thing to do with missing values ...

Missingness handling II

Code	Interpretation	Quantity
777	Not available or not applicable	0
888	Not done	?
999	Unkown	NA

Table 1: Missingness-quantity mapping

Three missingness codes were applied along *all the questionnaires/dataset*. For *economic analysis*, current mapping in table 1:

- if 777 (not available/applicable) this good/service is not provided, so its consumption is 0 by definition
- if the patient “said” 999 (which is “I know I used this resource, I don’t remember how much”) then this is a missing value
- 888 is blank completion. My current approach: for quantities this could plausibly be NA or actually 0, so any choice may be imperfect. Choose the less wrong, bearing in mind this is the end of a long demanding questionnaire for a frail population.

Missingness handling III

Some numbers in table 2.

	No 999	Has 999	All
No 888	206	19	225
Has 888	74	25	99
All	280	44	324

Table 2: CSRI missingness (codes 888 and 999) as of january 2026

So again (*especially* for CSRI/quantities¹):

- use 888 (not done) if the patient left blank
- use 777 (not available/applicable) if the patient left blank but the service/good is not done/available in your country
- use 999 (unknown) if the patients said “yes, I used it, but don’t know/remember how much”

¹eg FACT-G missingness type is less important, at least to me.

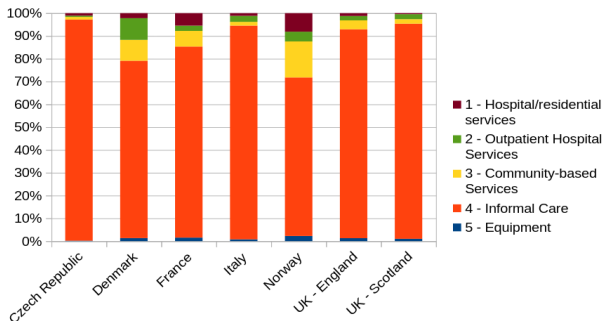
Some other general considerations regarding quantities/CSRI

Not much actually - mainly a TODO. To be mentioned:

- **hospital services** will be crosschecked against AE/hospitalizations days
- **medication** consumption is in progress: implementing a dictionary of active principles (if available) next time could be an improvement
- consumption associated to the experimental treatment has to be implemented

Consumption patterns

	Czech Republic	Denmark	France	Italy	Norway	UK - England	UK - Scotland
1 - Hospital/residential services	275.0	111.0	2056.0	491.0	523.0	521.0	37.0
2 - Outpatient Hospital Services	138.0	476.0	859.0	1100.0	274.0	821.0	179.0
3 - Community-based Services	285.0	458.0	2571.0	676.0	1014.0	1656.0	162.0
4 - Informal Care	24160.0	3896.0	31384.0	38984.0	4464.0	38824.0	7688.0
5 - Equipment	56.0	67.2	580.5	336.0	145.6	558.1	84.0



Summing all quantities (days for 1, visits for 2/3, hours for 4, months of usage for 5) table and graph above.

So, while hospitalisations/residential services are the main cost driver in monetary terms, wages (for informal care hours) dominate in quantity terms.

Part III

QOL/Utilities

So far we've talked only about unit cost and quantities ...

If we define:

mean_cost_T = mean of (unit costs · quantities) in experimental group

mean_cost_C = mean of (unit costs · quantities) in control group

Two similar statistics are of interest:

- Cost-utility analysis (primary)

$$\frac{\text{mean_cost}_T - \text{mean_cost}_C}{\text{mean_QALY}_T - \text{mean_QALY}_C}$$

eg: average cost per quality-adjusted life year (QALY) gained

- Cost-effectiveness analysis (secondary)

$$\frac{\text{mean_cost}_T - \text{mean_cost}_C}{\text{mean_FACTg}_T - \text{mean_FACTg}_C}$$

Utility basics

- definition
 - in *general*: measure of a certain person's **satisfaction** from a certain state of the world
 - in *our context*: *self evaluation* of satisfaction regarding health condition, normalized in range

$$u_t = \begin{cases} 1 & \text{perfect health} \\ 0 & \text{equivalent to death} \\ < 0 & \text{worse than being dead} \end{cases} \quad \text{at time } t$$

- sources: obtained from QOL questionnaires through mappings
 - EQ-5D-5L: direct mapping
 - FACT-G: indirect mapping applying Meregaglia et al. 2019

TODO: insert meregaglia equations?

Timepoint	NA	mean utility
Baseline	0	0.431522
W4	66	0.437461
W8	101	0.442907
W16	169	0.442532

Table 3: Mean raw utility, as of january 2026

From utility to QALY I

QALY (Quality Adjusted Life Years)

Assume you have the measurement of utility over time (measured in years): QALY is the area under this function.

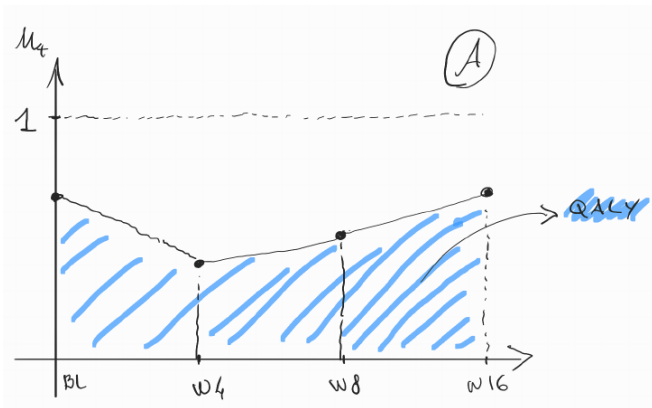
Examples (with constant utility)

A person who

- evaluates him/herself as *perfectly healthy* (utility=1) and is observed for 1 year, at the end of the year will have a $QALY = 1$ (area of $utility \cdot year = 1 \cdot 1 = 1$)
- has utility of 0.5, observed for 2 years will have a $QALY = 1$

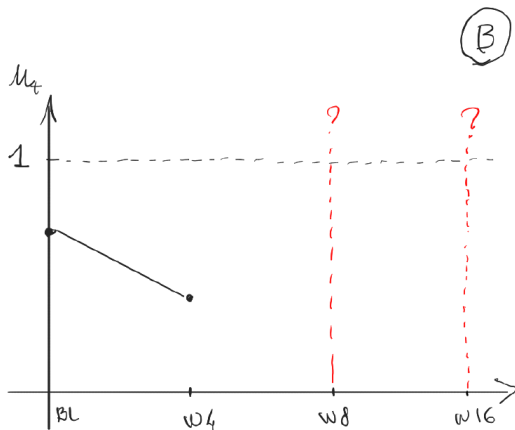
QALY I

When utility is not constant over time (as it often is) we can interpolate available observation/points. Eg wrt INSPIRE and full observations available



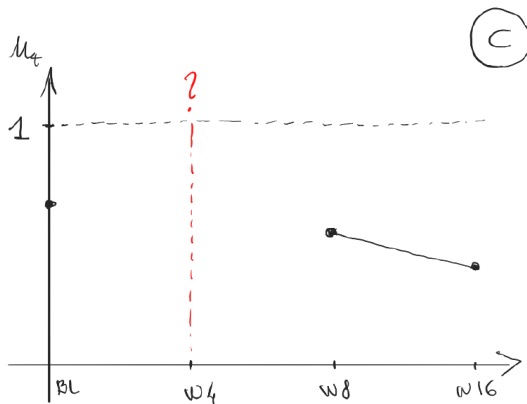
QALY II

No show at the end of follow up (first miss *after* last obs)



QALY III

No show during fup (first miss *before* last obs)



For what concerns INSPIRE:

- utility missingness patterns/profile (as of january) in table 4

missing pattern	count
A (no missing)	154
B (at the end only)	167
C (in between)	5
D (in btwn and at the end)	2

Table 4: Utility missingness pattern

- the theoretical maximum QALY a patient can have in our study (assuming he/she evaluates own status as “perfect health”) is $\approx$

$$1 \cdot \frac{16 \text{ weeks}}{52 \text{ weeks}} = 0.31$$

- as of january, QALYs can be calculated for 286 patients

QALY V

- 4 patients have values above 0.31 (check wrt to visit dates TODO)
- the algorithm implemented to obtain QALYs is still quite “draconian” but table 5 shows the distributions of a likely *underestimation* of final QALYs (due to follow up incompleteness)

	qaly
count	286
mean	0.111992
std	0.080735
min	0.005073
25%	0.058967
50%	0.118848
75%	0.146866
max	0.955801

Table 5: QALY distribution (jan 2026, underestimation)

Part IV

Next steps

Next steps

We have to:

- 1 prepare queries (up to january: not many)
- 2 complete data input (handle medication prices/quantities)
- 3 complete the analysis
- 4 write paper following Husereau et al. 2022

Thanks!

Questions?
(please set speed to 0.75x)

Part V

Misc

References I



Husereau, Don et al. (Jan. 2022). "Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 Explanation and Elaboration: A Report of the ISPOR CHEERS II Good Practices Task Force". en. In: *Value in Health* 25.1, pp. 10–31. ISSN: 10983015. DOI: 10.1016/j.jval.2021.10.008. URL: <https://linkinghub.elsevier.com/retrieve/pii/S1098301521017952> (visited on 07/31/2025).



Meregaglia, Michela et al. (Mar. 2019). "Mapping Health-Related Quality of Life Scores from FACT-G, FAACT, and FACIT-F onto Preference-Based EQ-5D-5L Utilities in Non-Small Cell Lung Cancer Cachexia". In: *The European Journal of Health Economics* 20.2, pp. 181–193. ISSN: 1618-7598, 1618-7601. DOI: 10.1007/s10198-017-0930-6. (Visited on 05/07/2026).